H.T.No: | | | | | | | | | | |

Course Code: **2501CS34**

# ADITYA UNIVERSITY

B. Tech – III Semester End Examinations Regular

## FUNDAMENTALS OF DATA SCIENCE

**(Common to CSE(DS))**

**Time: 3 hours** **Max. Marks: 100**

**1. Answer all 10 questions from Section-A. Each question carries 2 marks.**
**2. Answer one question from each unit in Section-B. Each question carries 16 marks.**

**SECTION –A** **10 X 2 =20**

| | | | | | |
|---|---|---|---|---|---|
| 1 | a | List the major stages of the Data Science Lifecycle | L1 | CO1 | [2M] |
| | b | Classify data into structured, semi-structured and unstructured data. | L3 | CO1 | [2M] |
| | c | Summarize any two methods for handling missing values. | L2 | CO2 | [2M] |
| | d | Differentiate between normalization and standardization.. | L2 | CO2 | [2M] |
| | e | Mention the purpose of Binary and HDF5 file formats. | L1 | CO3 | [2M] |
| | f | Differentiate between JSON, XML and HTML data formats. | L2 | CO3 | [2M] |
| | g | Differentiate between correlation and covariance. | L2 | CO4 | [2M] |
| | h | List any four data visualization techniques. | L1 | CO4 | [2M] |
| | i | Mention the use of NLTK in text mining. | L1 | CO5 | [2M] |
| | j | Differentiate between Neo4j and SQLite Database. | L2 | CO5 | [2M] |

**SECTION –B** **5 X 16 =80**

**UNIT-I**

| | | | | | |
|---|---|---|---|---|---|
| 2 | a | Define Data Science and explain its scope in various fields | L2 | CO1 | [8M] |
| | b | Describe the stages involved in the Data Science Lifecycle with examples. | L2 | CO1 | [8M] |
| | | **OR** | | | |
| 3 | a | Discuss the history and evolution of Data Science with major milestones | L2 | CO1 | [8M] |
| | b | List and explain the different tools and technologies commonly used in Data Science. | L2 | CO1 | [8M] |

**UNIT – II**

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|---|---|---|---|---|---|
| 4 | a | Discuss various data collection methods and give examples of how they are applied in real-world data science projects | L2 | CO2 | [8M] |
| | b | Describe data transformation techniques such as normalization and standardization with suitable examples. | L2 | CO2 | [8M] |
| | | **OR** | | | |
| 5 | a | Describe the process of data cleaning and explain different techniques used for handling missing values. | L2 | CO2 | [8M] |
| | b | Classify the steps involved in feature engineering and feature scaling, and review the role of data integration and aggregation in preprocessing. | L2 | CO2 | [8M] |

Model Question Paper

**UNIT – III**

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|---|---|---|---|---|---|
| 6 | a | Explain the process of reading and writing data in text format and summarize the steps involved in handling large text files in pieces. | L2 | CO3 | [8M] |
| | b | Demonstrate how JSON, XML, and HTML data formats are handled in Data Science, and review the techniques used in web scraping. | L2 | CO3 | [8M] |
| | | **OR** | | | |
| 7 | a | Distinguish between binary data formats such as HDF5 and traditional file formats. Indicate their advantages and suitable use cases | L2 | CO3 | [M] |
| | b | Discuss how to interact with relational databases and NoSQL systems like Mongo DB. Give examples of storing and retrieving data from each. | L2 | CO3 | [8M] |

**UNIT – IV**

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|---|---|---|---|---|---|
| 8 | a | Describe the importance of descriptive statistics in understanding data distribution. | L2 | CO4 | [8M] |
| | b | Discuss data visualization techniques and give examples of their applications in EDA. | L2 | CO4 | [8M] |
| | | **OR** | | | |
| 9 | a | Explain the role of correlation and covariance in identifying variable relationships. | L2 | CO4 | [8M] |
| | b | Classify univariate, bivariate, and multivariate analysis and indicate their use in anomaly detection. | L2 | CO4 | [M] |

**UNIT – V**

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|---|---|---|---|---|---|
| 10 | a | Describe the use of Neo4j in handling graph databases and give examples of its applications. | L2 | CO5 | [8M] |
| | b | Apply structure and functionality of Cypher, the graph query language used in Neo4j. | L3 | CO5 | [8M] |
| | | **OR** | | | |
| 11 | a | Discuss how the NLTK library is used for text mining and indicate its role in natural language processing. | L2 | CO5 | [8M] |
| | b | Apply different tools used for text analytics and review the use of SQLite in managing text-based data. | L3 | CO5 | [8M] |

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Model Question Paper